Module 5: Physical chemistry and transition elements

The content within this module assumes knowledge and understanding of the chemical concepts developed in Module 2: Foundations in chemistry and Module 3: Periodic table and energy.

This module extends the study of energy, reaction rates and equilibria, and the periodic table.

The main areas of physical chemistry studied include:

- rate equations, orders of reaction, the rate-determining step
- equilibrium constants, K_c and K_p
- acid–base equilibria including pH, K_a and buffer solutions
- lattice enthalpy and Born–Haber cycles
- entropy and free energy
- electrochemical cells.

The main areas of inorganic chemistry studied include:

- redox chemistry
- transition elements.

Synoptic assessment

This module provides a context for synoptic assessment and the subject content links strongly with the content encountered in Module 2: Foundations in chemistry and Module 3: Periodic table and energy.

- Atoms, moles and stoichiometry
- Acid and redox reactions
- Bonding and structure
- Periodicity, Group 2 and the halogens
- Enthalpy changes
- Reaction rates
- Chemical equilibrium

Knowledge and understanding of Module 2 and Module 3 will be assumed and examination questions will be set that link their content with this module and other areas of chemistry.

5.1 Rates, equilibrium and pH

The largely qualitative treatment of reaction rates and equilibria encountered in Module 3 is developed within a quantitative and graphical context.

This section also allows learners to develop practical quantitative techniques involved in the determination of reaction rates and pH.

There are many opportunities for developing mathematical skills, including use of logarithms and exponents, when studying the content of this section and when carrying out quantitative practical work.

5.1.1 How fast?

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
Orders, rate equations and rate constants		
(a)	explanation and use of the terms: *rate of reaction*, *order*, *overall order, rate constant*, *half-life*, *rate-determining step*	
(b)	deduction of: **(i)** orders from experimental data (ii) a rate equation from orders of the form: rate = $k[A]^m[B]^n$, where m and n are 0, 1 or 2	*M0.2* Learners are expected to interpret initial rates data to determine orders with respect to reactants. Integrated forms of rate equations are **not** required. **PAG10** HSW8 Use of rate equations.
(c)	calculation of the rate constant, k, and related quantities, from a rate equation including determination of units	*M0.0, M0.1, M0.4, M1.1, M2.2, M2.3, M2.4*
Rate graphs and orders		
(d)	from a concentration–time graph: **(i)** deduction of the order (0 or 1) with respect to a reactant from the shape of the graph (ii) calculation of reaction rates from the measurement of gradients **(see also 3.2.2 b)**	*M0.1, M0.4, M1.1, M3.1, M3.2, M3.3, M3.4, M3.5* Concentration–time graphs can be plotted from continuous measurements taken during the course of a reaction (continuous monitoring).
(e)	from a concentration–time graph of a first order reaction, measurement of constant half-life, $t_{1/2}$	*M3.1, M3.2* Learners should be aware of the constancy of half-life for a first order reaction.
(f)	for a first order reaction, determination of the rate constant, k, from the constant half-life, $t_{1/2}$, using the relationship: $k = \ln 2/t_{1/2}$	*M0.1, M0.4, M1.1, M2.3, M2.4, M2.5* Learners will **not** be required to derive this equation from the exponential relationship between concentration and time, $[A] = [A_0]e^{-kt}$.
(g)	from a rate–concentration graph: **(i)** deduction of the order (0, 1 or 2) with respect to a reactant from the shape of the graph (ii) determination of rate constant for a first order reaction from the gradient	*M0.1, M0.4, M1.1, M3.1, M3.2, M3.3, M3.4, M3.5* Rate–concentration data can be obtained from initial rates investigations of separate experiments using different concentrations of one of the reactants. Clock reactions are an approximation of this method where the time measured is such that the reaction has not proceeded too far. HSW5 Link between order and rate.

(h) the techniques and procedures used to investigate reaction rates by the initial rates method and by continuous monitoring, including use of colorimetry **(see also 3.2.2 e)**	**PAG9,10** HSW4 Opportunities to carry out experimental and investigative work.

Rate-determining step

(i) for a multi-step reaction, prediction of, **(i)** a rate equation that is consistent with the rate-determining step (ii) possible steps in a reaction mechanism from the rate equation and the balanced equation for the overall reaction	HSW1 Use of experimental evidence for the proposal of reaction mechanisms.

Effect of temperature on rate constants

(j) a qualitative explanation of the effect of temperature change on the rate of a reaction and hence the rate constant **(see 3.2.2 f–g)**	*M0.3*
(k) the Arrhenius equation: **(i)** the exponential relationship between the rate constant, k and temperature, T given by the Arrhenius equation, $k = Ae^{-Ea/RT}$ (ii) determination of E_a and A graphically using: $\ln k = -E_a/RT + \ln A$ derived from the Arrhenius equation.	*M0.1, M0.4, M2.2, M2.3, M2.4, M2.5, M3.1, M3.2, M3.3, M3.4* E_a = activation energy, A = pre-exponential factor, R = gas constant (provided on the *Data Sheet*) Explanation of A is **not** required. Equations provided on the *Data Sheet.* HSW5 Link between k and T.